BitFlip

CloudSherpa

Software Requirements Specification

Contents

1 Introduction

CloudSherpa is a multi-cloud cost management and optimization platform designed to address the growing complexity of managing cloud infrastructure across AWS, Azure, and Google Cloud Platform.

As organizations increasingly adopt multi-cloud strategies, they face challenges in monitoring spending, identifying cost anomalies, and optimizing resource utilization across fragmented platforms.

CloudSherpa consolidates billing and usage data from all three major cloud providers into a unified dashboard, providing real-time cost visibility, predictive forecasting, and actionable optimization recommendations.

2 User Stories / User Characteristics

2.1 User Characteristics

To adequately support CloudSherpa's multi-cloud optimization platform, the system targets three primary user groups.

- **Platform Administrator:**
  - *Characteristics:* High technical expertise, familiar with cloud infrastructure (AWS, Azure, GCP), IAM roles, and system administration.
  - *System Usage:* Responsible for configuring the platform, connecting/revoking cloud provider credentials and ensuring data ingestion pipelines are functioning.
- **Finance-Focused User:**
  - *Characteristics:* Low to medium technical expertise; high financial and business insight. Prioritizes plain-language summaries over technical jargon.
  - *System Usage:* Utilizes the platform to view forecasted cloud costs and track optimization recommendations for resources.

- **Technical Lead:**
  - *Characteristics:* High technical expertise; responsible for architecture, deployments, and infrastructure efficiency.
  - *System Usage:* Monitors cross-provider resource usage, identifies idle resources, and applies right-sizing recommendations.

2.2 User Stories

2.2.1 Platform Administrator

- **US1:** As a platform user, I want to connect my AWS, Azure, and GCP accounts so that CloudSherpa can collect usage, billing, and performance data.
- **US2:** As a platform administrator, I want provider-specific ingestion jobs to run automatically so that data remains up to date without manual effort.
- **US3:** As a platform user, I want failed cloud integrations to show clear error feedback so that I can fix configuration issues quickly.
- **US4:** As a system administrator, I want processed data pipelines to be reliable so that dashboards and services reflect trusted information.
- **US5:** As a new user, I want to create an account and log in securely so that I can access CloudSherpa safely.
- **US6:** As a platform user, I want to link multiple cloud environments to my profile so that I can manage them from one place.
- **US7:** As a platform user, I want to manage and revoke connected cloud accounts so that I stay in control of my integrations.

2.2.2 Finance-Focused User

- **US8:** As a finance-focused user, I want forecasted cloud costs for upcoming periods so that I can plan budgets proactively.
- **US9:** As a platform user, I want to see spending trends over time so that I can identify whether my costs are rising or falling.
- **US10:** As a non-technical user, I want the dashboard to present information clearly so that I can still make informed decisions.

- **US11:** As a platform user, I want trend analysis summaries so that I can understand long-term cost behavior.
- **US12:** As an auditor or reviewer, I want visual summaries of key metrics so that I can assess cloud usage efficiently.

2.2.3 Technical Lead

- **US13:** As an operations manager, I want forecasts per provider so that I can understand where future spend will be concentrated.
- **US14:** As a platform user, I want the system to identify idle resources so that I can reduce unnecessary cloud spend.
- **US15:** As a platform user, I want recommendations for overprovisioned resources so that I can right-size infrastructure.
- **US16:** As a platform user, I want recommendations to be grouped by severity so that I can focus on the highest-value optimizations first.
- **US17:** As a platform user, I want a dashboard overview of cloud usage and spend so that I can understand my environment at a glance.
- **US18:** As a platform user, I want visual charts for forecasts and trends so that I can interpret data quickly.

3 Use Cases

3.1 High-Level Use Case Diagram

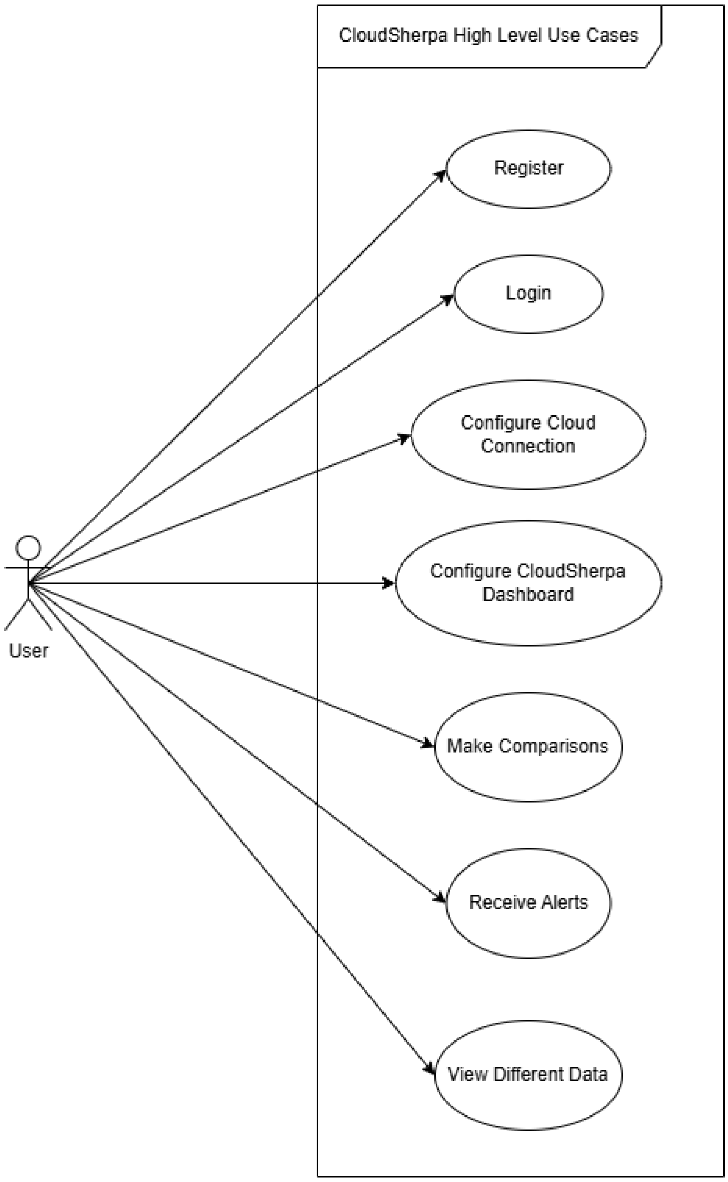

Figure 1: High-Level Use Case Diagram

3.2 Use Cases

Use Case Conventions & Methodology

Terminology & Abbreviations:
- **TUCBW:** ”This use case begins with...”
- **TUCEW:** ”This use case ends with...”

Documentation Methodology:
To ensure clarity while avoiding unnecessary document bloat, this section utilizes two distinct levels of abstraction based on the complexity of the business process:
- **High-Level Use Cases:** Utilized for less complicated operations. These specify exactly when and where the use case begins and ends without detailing background processing.
- **Expanded Use Cases:** Utilized for complex operations. These provide a detailed, step-by-step primary scenario of how the actor interacts with the system to carry out the task.

- **1. Authentication**
  - **1.1 Register:**
    TUCBW a new user navigates to the login page and submits the registration form. TUCEW the user is redirected to the dashboard page as a logged-in user.

  - **1.2 Login:**
    TUCBW an existing user submits their credentials on the login page. TUCEW the user is authenticated and redirected to the dashboard.

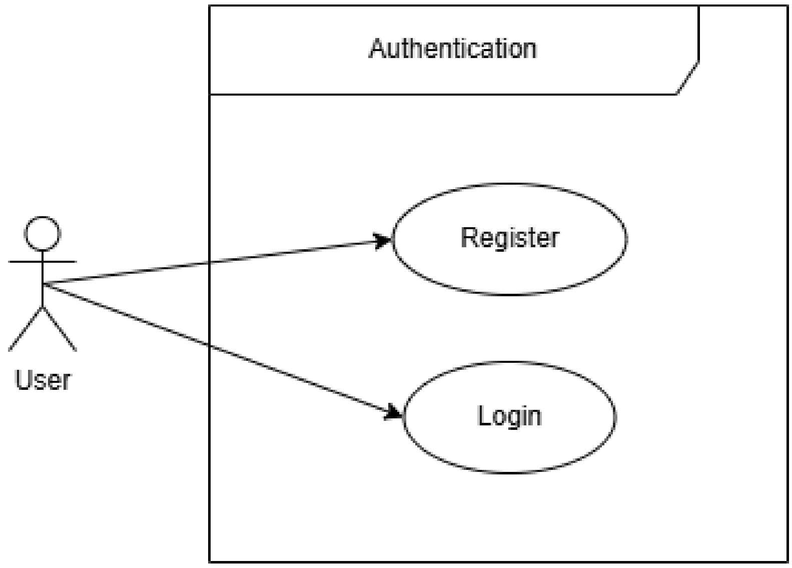

Figure 2: Authentication

- **2. Themes**
  - **2.1 Switch to dark mode:**
    TUCBW the user is logged in with the theme set to light mode. The user clicks the 'Dark Mode' button. TUCEW the user sees the theme change to dark mode.
  - **2.2 Switch to light mode:**
    TUCBW the user is logged in with the theme set to dark mode.. TUCEW the user sees the theme change to light mode.

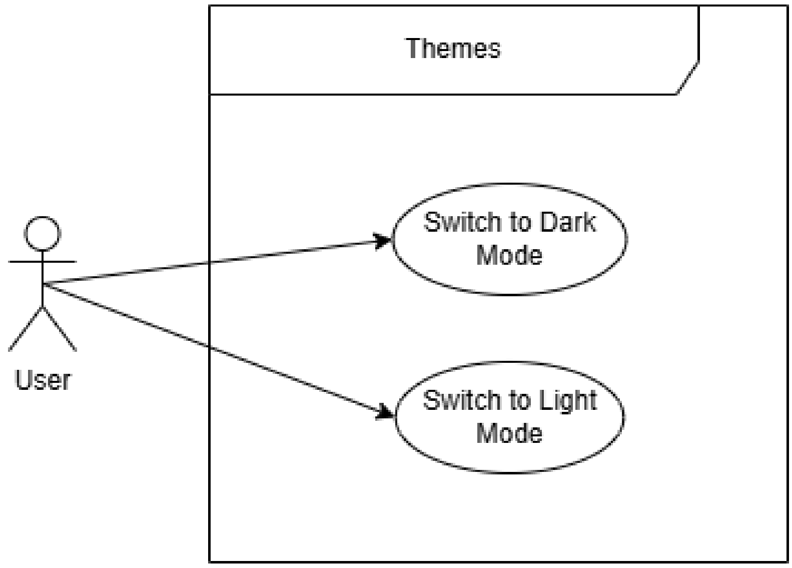

Figure 3: Themes

- **3. Dashboard**
  - **3.1 Add Widget:**
    TUCBW the user clicks 'Add Chart' or 'Add KPI' while in the dashboard 'Edit mode'.
    TUCEW the user sees a newly configured widget on the dashboard layout.
    * **ID:** UC3.1.1
    * **Name:** Add Line Chart
    * **Actor:** User
    * **Precondition:** The user has added an unconfigured chart widget to the layout and has saved their changes, exiting dashboard 'Edit mode'..
    * **Primary Scenario:**
      1. TUCBW the User clicks the 'Configure Widget' button on the unconfigured chart widget.
      2. The system prompts the user to enter a display name and select a chart type.
      3. The User enters a display name and selects 'Line Chart' from the chart type dropdown.
      4. The system updates the configuration form to display line chart-specific metric fields.

5. The User selects the desired metrics and clicks the ’Save’ button.
6. The system verifies the configuration parameters and renders the active line chart on the new widget.

- **ID:** UC3.1.2
- **Name:** Add Guage Chart
- **Actor:** User
- **Precondition:** The user has added an unconfigured chart widget to the layout and has saved their changes, exiting dashboard ’Edit mode’.
- **Primary Scenario:**
  1. TUCBW the User clicks the ’Configure Widget’ button on the unconfigured chart widget.
  2. The system prompts the user to enter a display name and select a chart type.
  3. The User enters a display name and selects ’Gauge Chart’ from the chart type dropdown.
  4. The system updates the configuration form to display line chart-specific metric fields.
  5. The User selects the desired metrics and clicks the ’Save’ button.
  6. The system verifies the configuration parameters and renders the active Gauge chart on the dashboard.
- **ID:** UC3.1.3
- **Name:** KPI Indicator
- **Actor:** User
- **Precondition:** The user has added an unconfigured KPI widget to the layout and has saved their changes, exiting dashboard ’Edit mode’.
- **Primary Scenario:**
  1. TUCBW the User clicks the ’Configure Widget’ button on the unconfigured KPI widget.
  2. The system prompts the user to configure the data that should be tracked.
  3. The User clicks the ’Save’ button.

4. The system verifies the configuration and displays a KPI indicator.

- **3.2 Edit Layout:**
  TUCBW the user is on a preconfigured dashboard with existing widgets and clicks the pencil icon to start edting. The user resizes widgets and drags them to the desired positions. TUCEW the user saves the dashboard and sees the widgets in their new positions.
- **3.3 Configure Mutliple Layouts:**
  TUCBW the user opens the dashboard selector and selects the option to create a new dashboard. TUCEW the user sees the page refresh to show a blank layout with the new dashboard name displayed in the dashboard selector.
- **3.4 Select Dashboard Layout**
  TUCBW the user opens the dashboard selector dropdown and sees a list of preconfigured dashboard to select from. TUCEW the user sees the page refresh to display the name of the selected dashboard in the dashboard dropdown and the layout of the selected dashboard.
- **3.5 Adjust Temporal Window:**
  TUCBW the user is on the dashboard page with a preconfigured dashboard. TUCEW the user sees the chart widgets adjust to display data from relating to the selected time window.
- **3.6 Reflect Live Metrics:**
  TUCBW the user opens a preconfigured dashboard with chart widget setup to display metrics for a resource. TUCEW the user sees the data being displayed on the chart widgets update after a period of time.

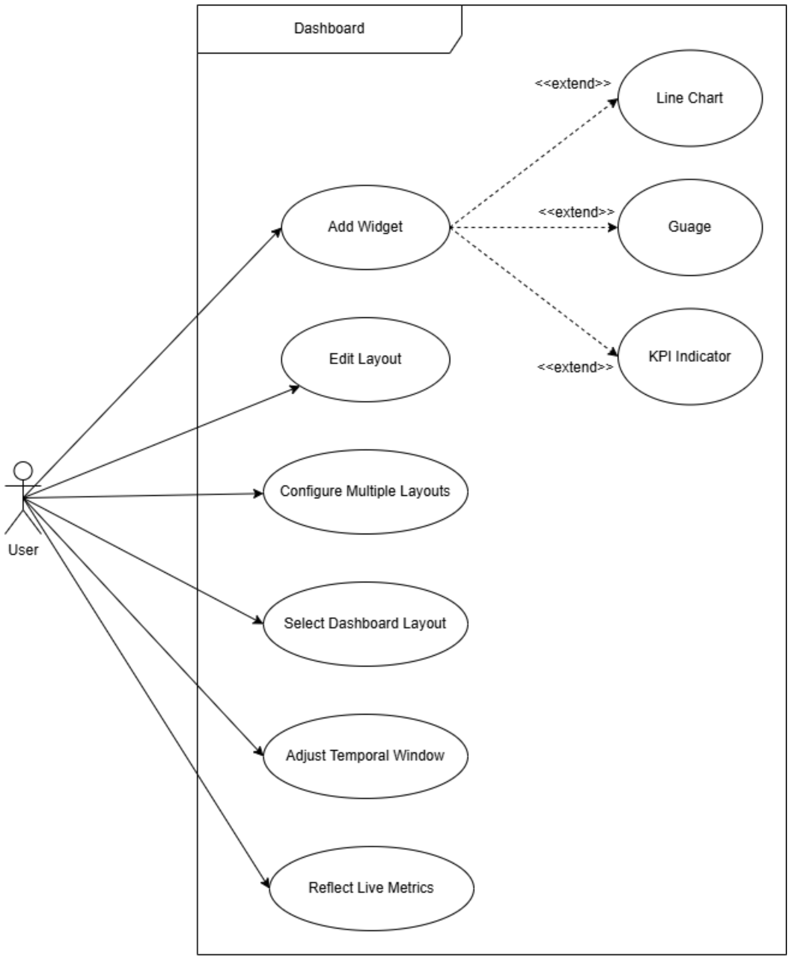

Figure 4: Dashboard

- **4 KPI WIdget**
  - **4.1 Add Widget:**
    TUCBW the user clicks 'Add KPI' while in the dashboard 'Edit mode'.
    TUCEW the user sees the newly configured widget on the dashboard layout.
    - **ID:** UC4.1.1
    - **Name:** Configure Billing Changes to Monitor
    - **Actor:** User
    - **Precondition:** The user has added an unconfigured chart widget to the layout and has saved their changes, exiting dashboard 'Edit mode'.
    - **Primary Scenario:**

1. TUCBW the User clicks the ’Configure Widget’ button on the unconfigured KPI widget.
2. The user selects a resource whose costs should be aggregated.
3. The user clicks ’Save’
4. The the user gets redirected to the dashboard page
5. The system verifies the configuration parameters and renders the kpi indicator.

- **ID:** UC4.1.2
- **Name:** Choose Billing Aggregation period
- **Actor:** User
- **Precondition:** The user has added an unconfigured chart widget to the layout and has saved their changes, exiting dashboard ’Edit mode’.
- **Primary Scenario:**
  1. TUCBW the User clicks the ’Configure Widget’ button on the unconfigured KPI widget.
  2. The user configures the rest of the widget parameters.
  3. The user selects a time period for which the costs should be aggregated from the time period dropdown.
  4. The user clicks ’Save’
  5. The the user gets redirected to the dashboard page
  6. The system verifies the configuration parameters and renders the kpi indicator displaying the data that is based on the time period selected.
- **ID:** UC4.1.3
- **Name:** Change KPI Display Title
- **Actor:** User
- **Precondition:** The user has added an unconfigured chart widget to the layout and has saved their changes, exiting dashboard ’Edit mode’.
- **Primary Scenario:**

1. TUCBW the User clicks the 'Configure Widget' button on the unconfigured KPI widget.
2. The sees the input that is filled with the default 'New KPI' widget name.
3. The user updates the input to contain the desired display name.
4. The user clicks 'Save'
5. The the user gets redirected to the dashboard page
6. The system verifies the configuration parameters and renders the kpi indicator with the new KPI widget display name .

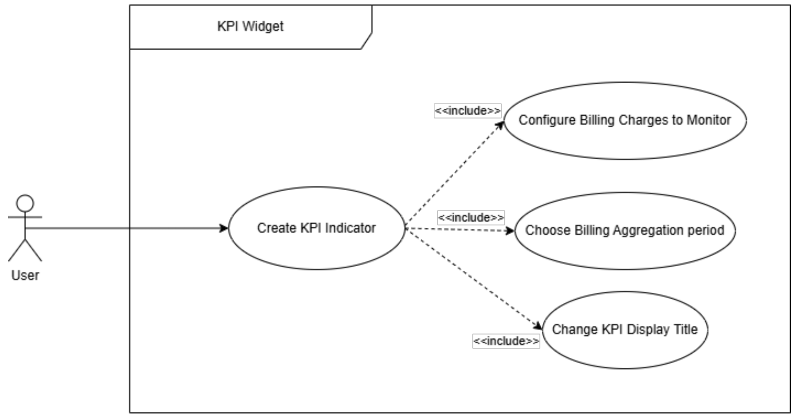

Figure 5: KPI Widget

– **5. Configure Cloud Connection**

* **5.1 Add Cloud Connection:**
TUCBW the user is logged in.
TUCEW the user sees the new connection in the connection manager.

  · **ID:** UC5.1.1

  · **Name:** Add AWS Connection

  · **Actor:** User

  · **Precondition:** The user is logged in.

  · **Primary Scenario:**

1. TUCBW the User clicks the 'AWS' link under the 'Add Connection' category on the sidebar and gets redirected to the AWS setupWizard.

2. The user sees the step 1 and fills out the required details and clicks ’Next’.
3. The User sees the step 1 form get replaced with the step 2 form.
4. The User selects the cloud services they wish to monitor and clicks ’next’.
5. The system displays a progress bar representing the queue for the service discovery and displays the step 3 form once service discovery has concluded.
6. The User selects the resources they wish to monitor on the dashboard and clicks ’Finish’.
7. The system persists the connection and redirects the User to the connection manager page
8. The user sees the newly created AWS connection card populated in the connection manager page.

- **ID:** UC5.1.1
- **Name:** Add GCP Connection
- **Actor:** User
- **Precondition:** The user is logged in.
- **Primary Scenario:**

1. TUCBW the User clicks the ’GCP’ link under the ’Add Connection’ category on the sidebar and gets redirected to the GCP setupWizard.
2. The user sees the step 1 and fills out the required details and clicks ’Next’.
3. The User sees the step 1 form get replaced with the step 2 form.
4. The User selects the cloud services they wish to monitor and clicks ’next’.
5. The system displays a progress bar representing the queue for the service discovery and displays the step 3 form once service discovery has concluded.
6. The User selects the resources they wish to monitor on the dashboard and clicks ’Finish’.

7. The system persists the connection and redirects the User to the connection manager page
8. The user sees the newly created GCP connection card populated in the connection manager page.

- **ID:** UC5.1.1
- **Name:** Add Azure Connection
- **Actor:** User
- **Precondition:** The user is logged in.
- **Primary Scenario:**

1. TUCBW the User clicks the ’Azure’ link under the ’Add Connection’ category on the sidebar and gets redirected to the Azure setupWizard.
2. The user sees the step 1 and fills out the required details and clicks ’Next’.
3. The User sees the step 1 form get replaced with the step 2 form.
4. The User selects the cloud services they wish to monitor and clicks ’next’.
5. The system displays a progress bar representing the queue for the service discovery and displays the step 3 form once service discovery has concluded.
6. The User selects the resources they wish to monitor on the dashboard and clicks ’Finish’.
7. The system persists the connection and redirects the User to the connection manager page
8. The user sees the newly created Azure connection card populated in the connection manager page.

* **5.2 Disable Cloud Connection:**
TUCBW the logged in User is on the Connection Manager page with a preconfigured connection and clicks on the options button for the connection and selects ’View Resources’ where they disable the resources for the specified connection. TUCEW the User sees that they can not see the disabled resources when configuring the widgets as an option to monitor.

* **5.3 View Cloud Connection Health:**
  TUCBW the logged in User is on the Connection Manager page with a preconfigured connection. TUCEW the user sees a badge in the bottom right corner of the connection card indicating the status of the connection.

* **5.4 Remove Cloud Connection:**
  TUCBW the logged in User is on the Connection Manager page with a preconfigured connection and clicks the delete button with the 'trash' icon and prompted to confirm the deletion. TUCEW the page rerenders and the User sees the connection is removed and a toast notification confirming the deletion

* **5.5 Receive failed connection notification:** TUCBW the logged in User clicked 'finish' after completing the setup wizard for a specified connection. TUCEW the User sees a toast notification that the operation for establishing a connection failed if the connection could not be created.

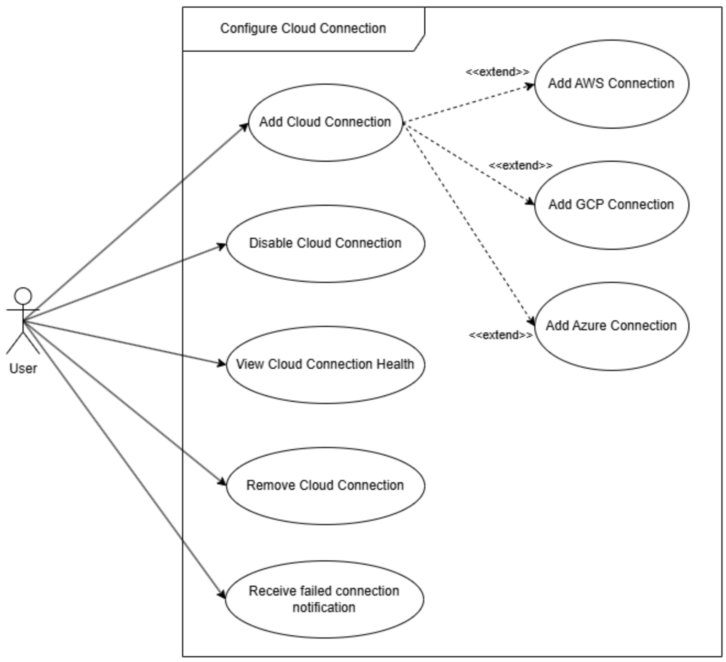

Figure 6: Configure Cloud Connections

– **6. Add Connection**

* **6.1 Configure CloudSherpa IAM User:**
TUCBW the logged in User is on step 2 of the setup wizard they see the provided configuration required to monitor those the desired services on CloudSherpa and Add it to their Cloud Account. TUCEW the User succesfully creates the cloud connection and sees the data from their account being ingested on the dashboard.

* **6.2 Select Instances CloudSherpa Should Monitor:**
TUCBW the logged in User is on step 2 of the setup wizard and selects the services they want to monitor from their cloud account and clicks 'Next'. TUCEW the User is taken to step 3 and sees the resources connected to the selected services of their cloud account.

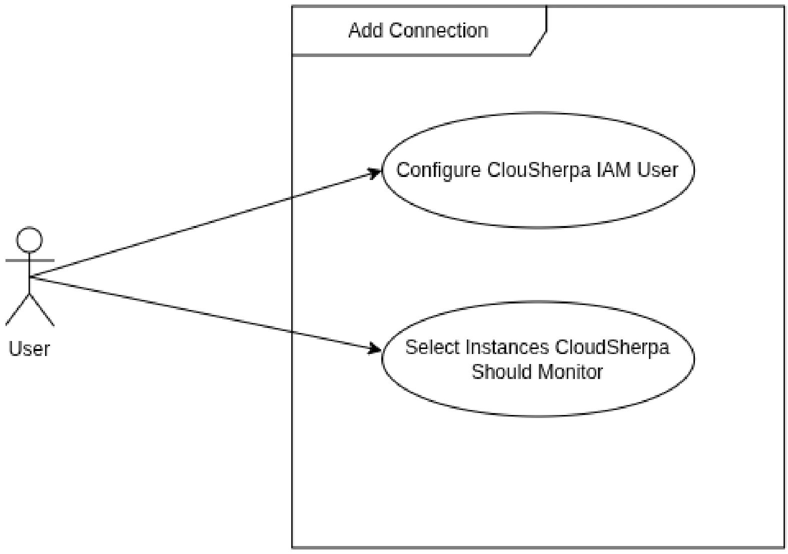

Figure 7: Add AWS Connection

- **7. Manage Connections**

  * **7.1 Add Cloud Connection:**
  TUCBW the logged in User is on the Connection Manager page and clicks the 'Add' button and gets redirected to the setup wizard for the provider they chose from the dropdown. TUCEW the User sees the connection populate on the Connection Manager page.

  * **7.2 Remove Cloud Connection:**
  TUCBW the logged in User is on the Connection Manager page with preconfigured connections and clicks the delete button with the 'trash' icon on the desired connection. TUCEW the user sees a toast notification confirming the deletion and the connection removed from the

connection manager page.

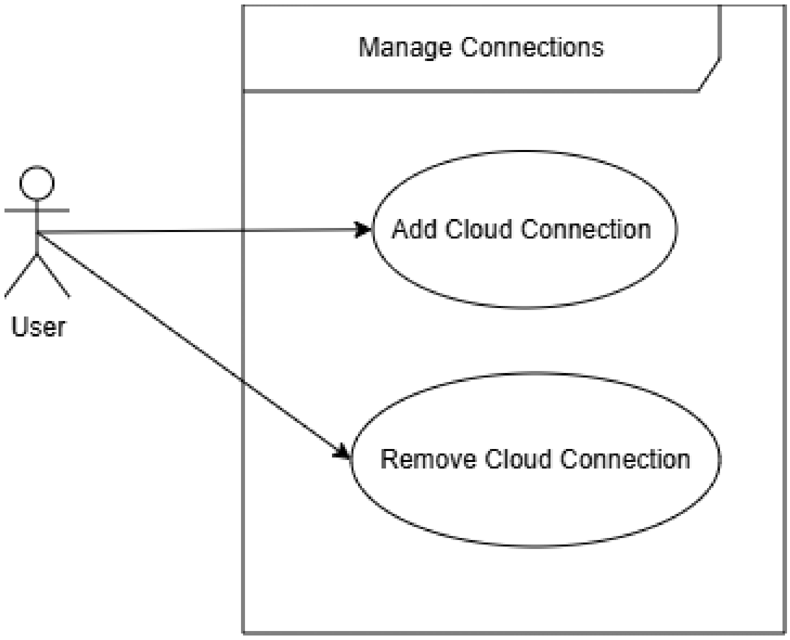

Figure 8: Manage Connections

– **8. Resources**

  * **8.1 Receive Idle Resource Notification:**
    TUCBW the logged in User is on the dashboard with a preconfigured layout. TUCEW the User sees a warning icon on widgets that have no data to display

  * **8.2 REceive Recommendations for Redundant Resources:**
    TUCBW the logged in User is on the dashboard with a preconfigured layout. TUCEW the user sees a 'sparkles' icon on widgets whose related resources have optimization recommendations.

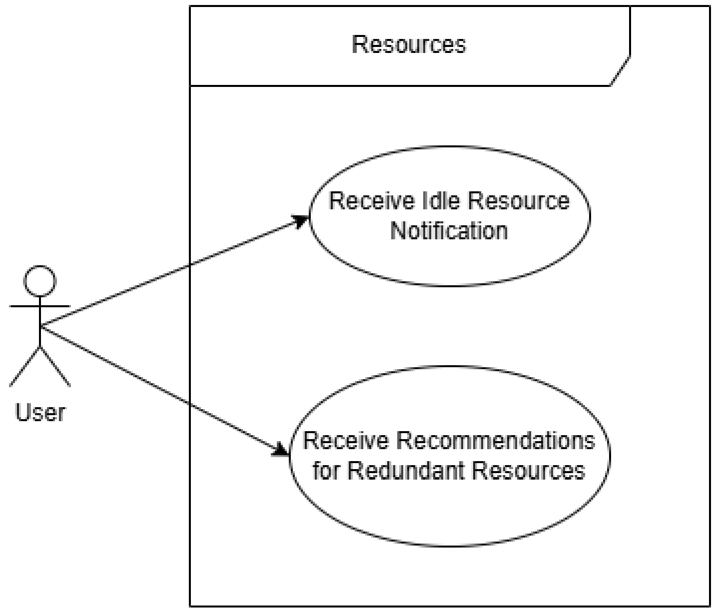

Figure 9: Resources

- **9. Manage Resources**
  - **9.1 Add Resource for Monitoring**
    TUCBW the logged in User navigates to the Resource Manager page for the desired connection and enables the desired resource by toggling the related switch on. TUCEW the user sees the newly enabled resource displayed as an option for monitoring in the widget configuration forms.
  - **9.2 Remove Resource form Monitoring**
    TUCBW the logged in User navigates to the Resource Manager page for the desired connection and disabled the desired resource by toggling the related switch off. TUCEW the user sees the disabled resource is not displayed as an option for monitoring in the widget configuration forms.

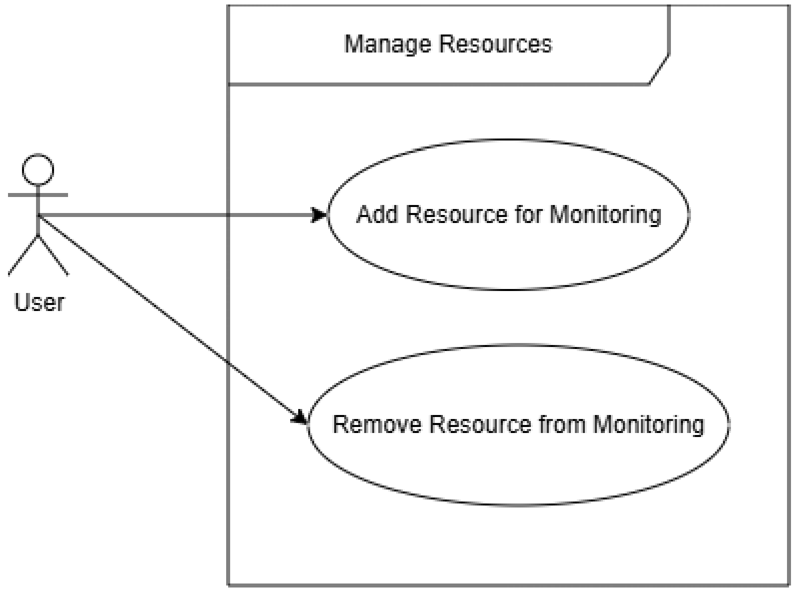

Figure 10: Manage Resources

- **10. Historical Cloud Data:**
  * **10.1 Analyze Historical Data:**
    TUCBW the user is logged in and is on the dashboard page with a dashboard selected.
    TUCEW the user sees historical data displayed on the widgets.
    - **ID:** UC10.1.1
    - **Name:** Analyze Historical Cost
    - **Actor:** User
    - **Precondition:** The logged in User is on the dashboard with a dashboard layout selected.
    - **Primary Scenario:**
    1. The User adds a KPI widget to the dashboard.
    2. The User saves the new layout.
    3. The User configures the the KPI widget to track the cost for the desired resources over the desired time period and saves the configuration.
    4. The User gets redirected to the dashboard page.

5. The system verifies the configuration and populates the widget with the historical data relating to the configuration
6. The user sees the data on the widget and uses it to analyze the historical cost.

- **ID:** UC10.1.2
- **Name:** Analyze Historical Usage
- **Actor:** User
- **Precondition:** The logged in User is on the dashboard with a dashboard layout selected.
- **Primary Scenario:**

1. The User adds a Chart widget to the dashboard.
2. The User saves the new layout.
3. The User configures the the Chart widget to track the usage metrics for the desired resources and saves the configuration.
4. The User gets redirected to the dashboard page.
5. The system verifies the configuration and populates the widget with the historical data relating to the configuration for the time period selected for the dashboard.
6. The user sees the data on the widget and uses it to analyze the historical usage metrics.

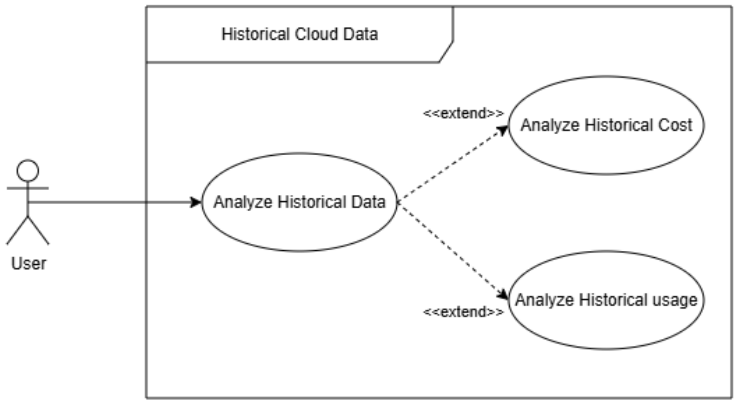

Figure 11: Historical Cloud Data

- **11. Recent Ingested Cloud Data**
  - **11.1 Configure Ingestion Period**
    - **ID:** UC11.1
    - **Name:** Configure Ingestion Period
    - **Actor:** User
    - **Precondition:** The user is logged and currently busy with step 3 of a setup wizard.
    - **Primary Scenario:**

    1. TUCBW the User uses the slider to choose an ingestion period to select how often new data should be ingested.
    2. The user clicks ’finish’
    3. The system saves the ingestion period along with the connection configuration.
    4. TUCEW the user sees the saved connection and its ingestion period on the Connection Manager page.
  - **11.1 Analyze Recent Data**
    - **ID:** UC11.2
    - **Name:** Analyze Recent Data
    - **Actor:** System / User
    - **Precondition:** An ingestion period has been configured and saved.
    - **Primary Scenario:**

    1. TUCBW the system retrieves the cloud data periodically based on the ingestion period.
    2. TUCEW The system aggregates the base metrics and presents the recent data to the User on their dashboards.
  - **11.1.1 Analyze Recent Cost**
    - **ID:** UC11.1.1
    - **Name:** Analyze Recent Cost
    - **Actor:** User

- **Precondition:** The system has ingested the cloud data and is displaying it on the Users configured dashboards.
- **Primary Scenario:**

1. TUCBW the User is on the dashboard with preconfigured KPI widgets.
2. TUCEW the User sees the data on the KPI widgets update when new data is ingested.

* **11.1.2 Analyze Recent Cost**

- **ID:** UC11.1.2
- **Name:** Analyze Recent Usage
- **Actor:** User
- **Precondition:** The system has ingested the cloud data and is displaying it on the Users configured dashboards.
- **Primary Scenario:**

1. TUCBW the User is on the dashboard with preconfigured Chart widgets.
2. TUCEW the User sees the data on the chart widgets update when new data is ingested.

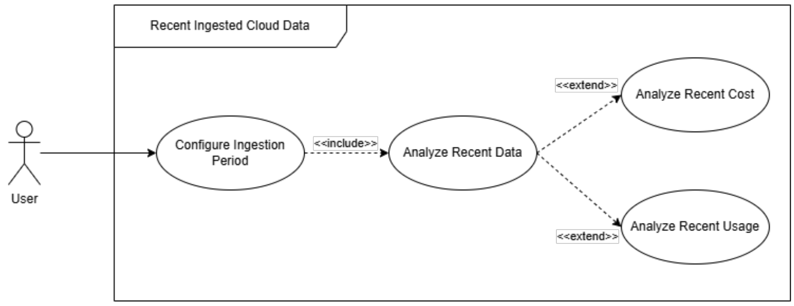

Figure 12: Recent Ingested Cloud Data

– **12. Time-Series Data**

– **12.1 View Time-Series Data**
TUCBW the User is logged in and viewing a dashboard layout with chart widgets setup.
TUCEW the User sees the time series data on the Chart widgets.

- **12.1.1 View Daily Data**

  - **ID:** UC12.1.1
  - **Name:** View Daily Data
  - **Actor:** User
  - **Precondition:** The system has ingested the cloud data and the User is on the dashboard page viewing a layout with configurd Chart widgets.
  - **Primary Scenario:**

  1. TUCBW the User clicks the 'Time Period' selector on the dashboard and selects '24 hours'.
  2. The system saves the time period and updates the Chart widgets to only display data that falls in the specified time period.
  3. TUCEW the User sees the Chart widgets display the data for the last 24 hours.

- **12.1.2 View Daily Data**
  - **ID:** UC12.1.2
  - **Name:** View Daily Data
  - **Actor:** User
  - **Precondition:** The system has ingested the cloud data and the User is on the dashboard page viewing a layout with configured Chart widgets.
  - **Primary Scenario:**

  1. TUCBW the User clicks the 'Time Period' selector on the dashboard and selects '24 hours'.
  2. The system saves the time period and updates the Chart widgets to only display data that falls in the specified time period.
  3. TUCEW the User sees the Chart widgets display the data for the last 24 hours.

- **12.1.3 View Weekly Data**
  - **ID:** UC12.1.3

- **Name:** View Weekly Data
- **Actor:** User
- **Precondition:** The system has ingested the cloud data and the User is on the dashboard page viewing a layout with configured Chart widgets.
- **Primary Scenario:**

1. TUCBW the User clicks the 'Time Period' selector on the dashboard and selects '7 days'.
2. The system saves the time period and updates the Chart widgets to only display data that falls in the specified time period.
3. TUCEW the User sees the Chart widgets display the data for the last 7 days.

* **12.1.3 View Monthly Data**

- **ID:** UC12.1.3
- **Name:** View Monthly Data
- **Actor:** User
- **Precondition:** The system has ingested the cloud data and the User is on the dashboard page viewing a layout with configured Chart widgets.
- **Primary Scenario:**

1. TUCBW the User clicks the 'Time Period' selector on the dashboard and selects '30 days'.
2. The system saves the time period and updates the Chart widgets to only display data that falls in the specified time period.
3. TUCEW the User sees the Chart widgets display the data for the last 30 days.

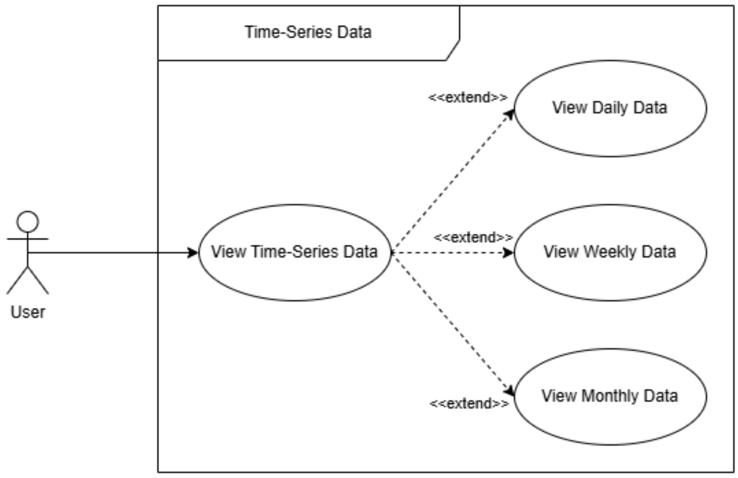

Figure 13: Time-Series Data

– **13. Future Cloud Cost**

  * **13.1 Request Forecasts:**
    TUCBW the logged in User is on a 'Forecasts' page and uses the drop-downs to configure the forecasts. TUCEW the User sees the historical data as well as the forecasts for that desired data populate the page.

  * **13.2 View Forecasts**
    TUCBW the User has configured the Usage Forecasts page to display the forecasts for the desired resource metrics
    TUCEW the User sees the forecasts for the desired Usage

    · **13.1.1 View GCP Forecasts**

    · **ID:** UC13.1.1

    · **Name:** View GCP Forecasts

    · **Actor:** User

    · **Precondition:** The User is on the Usage Forecasts page with AWS cloud connections already setup.

    · **Primary Scenario:**

    1. TUCBW the User selects 'GCP' as the provider in the configuration menu.

    2. the User selects the desired resources and metrics to view.

3. TUCEW the User sees the chart and summary cards on the Usage Forecasts page populate with the forecasts for the selected ’GCP’ connection resources.

- **13.1.2 View AWS Forecasts**
- **ID:** UC13.1.2
- **Name:** View AWS Forecasts
- **Actor:** User
- **Precondition:** The User is on the Usage Forecasts page with AWS cloud connections already setup.
- **Primary Scenario:**

1. TUCBW the User selects ’AWS’ as the provider in the configuration menu.
2. the User selects the desired resources and metrics to view.
3. TUCEW the User sees the chart and summary cards on the Usage Forecasts page populate with the forecasts for the selected ’AWS’ connection resources.

- **13.1.3 View Azure Forecasts**
- **ID:** UC13.1.3
- **Name:** View Azure Forecasts
- **Actor:** User
- **Precondition:** The User is on the Usage Forecasts page with Azure cloud connections already setup.
- **Primary Scenario:**

1. TUCBW the User selects ’Azure’ as the provider in the configuration menu.
2. the User selects the desired resources and metrics to view.
3. TUCEW the User sees the chart and summary cards on the Usage Forecasts page populate with the forecasts for the selected ’Azure’ connection resources.

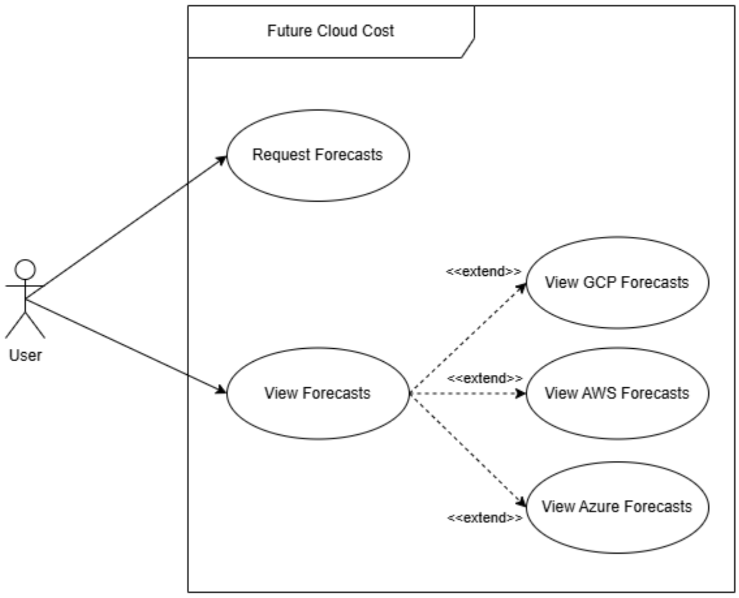

Figure 14: Future Cloud Cost

- **14. Recommendations and Anomolies**
  - **14.1 View Recommendations On Dashboard**
    TUCBW the logged in User is viewing a preconfigured dashboard with widgets already configured
    TUCEW the User sees a 'sparkles' icon on the on the widgets that are displaying data from a resource that has a rightsizing recommendation related to it.
    - **14.1.1 View Optimization Recommendations by Severity**
    - **ID:** UC14.1.1
    - **Name:** View Optimization Recommendations by Severity
    - **Actor:** User
    - **Precondition:** The user is viewing a preconfigured dashboard with widgets already configured and sees the 'sparkles' icon on a widget that indicates the resource has a recommendation related to it.
    - **Primary Scenario:**

    1. TUCBW the User clicks on the 'sparkles' icon button and gets redirected to the recommendations page and sees the recommendation related to the widget resource open up.

2. TUCEW the User sees the recommended action related to the resource as well as the evidence that justifys the recommendation.

* **14.2 Receive New Optimization Oppurtunity Notification**
  TUCBW the logged in User is viewing a preconfigured dashboard with widgets already configured
  TUCEW the User sees a 'sparkles' icon on the on the widgets that are displaying data from a resource that has a rightsizing recommendation related to it.

* **14.3 View Anomalies on Dashboard**
  TUCBW the logged in User is viewing a preconfigured dashboard with widgets already configured
  TUCEW the User sees a 'sparkles' icon on the on the widgets that are displaying data from a resource that has a rightsizing recommendation related to it.

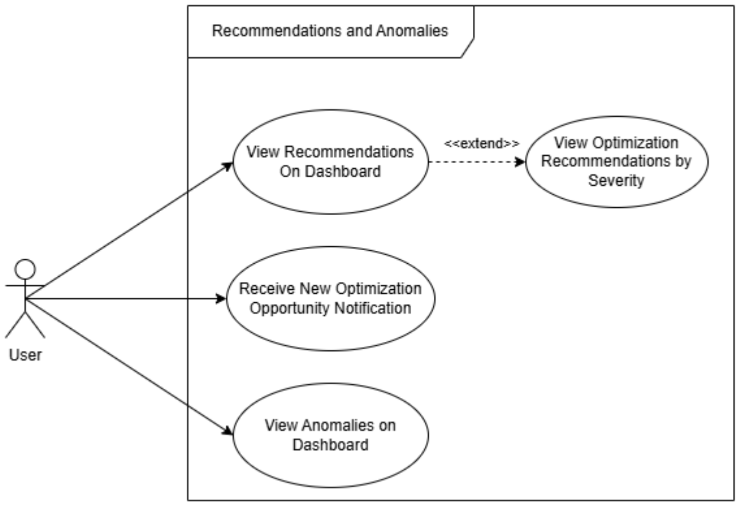

Figure 15: Recommendations and Anomalies

– **15. Recommendations and Anomolies**

* **15.1 View Analytics KPI's**
  TUCBW the User is on the Dashboard page with connection configured to include billing data already setup and adds KPI widgets.
  TUCEW the User sees the Billing Summaries on the KPI widgets.

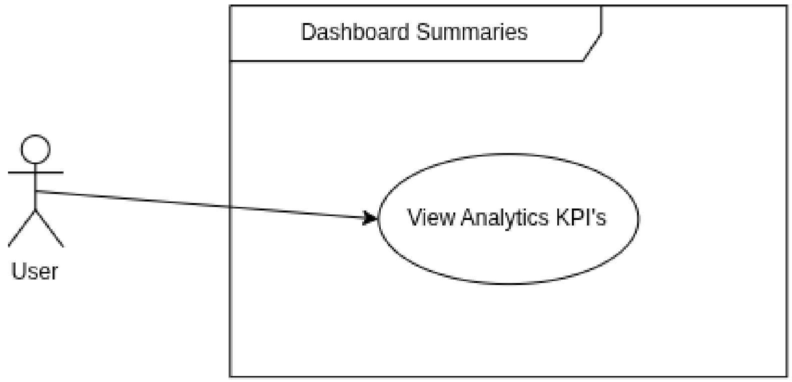

Figure 16: Billing Summaries

4 Functional Requirements

4.1 Cloud Data Integration

FR1: Cloud Data Integration *(Subsystem: Ingestion Service - Connection Manager and Cloud Connector Modules)*

- **FR1.1:** The system shall integrate with major cloud providers (AWS, Azure, and GCP) via their native APIs and cost reporting tools.
  - **FR1.1.1** The system shall integrate with the AWS Cloud Provider.
    - **FR1.1.1.1** The system shall ingest usage metrics via AWS CloudWatch.
    - **FR1.1.1.2** The system shall ingest billing data via AWS Cost and Usage Reports (CUR).
  - **FR1.1.2** The system shall integrate with the Microsoft Azure Cloud Provider.
    - **FR1.1.2.1** The system shall ingest metrics via Azure Monitor.
    - **FR1.1.2.2** The system shall ingest billing data via Azure Cost Management APIs.
  - **FR1.1.3** The system shall integrate with the Google Cloud Platform (GCP).
    - **FR1.1.3.1** The system shall ingest metrics via GCP Cloud Monitoring.
    - **FR1.1.3.2** The system shall ingest billing data via GCP Cloud Billing.

- **FR1.2:** The system shall support configurable ingestion frequencies (e.g., hourly, daily, and monthly synchronization).

4.2 Data Storage and Processing

FR2: Data Storage and Processing *(Subsystem: Ingestion Service - Normalization module and Persistence Service)*

- **FR2.1:** The system shall store historical cloud data.
- **FR2.2:** The system shall process, normalize, and aggregate time-series data to prepare it for analytical workloads.
  - **FR2.2.1** The system shall normalize terminology across cloud providers.
  - **FR2.2.2** The system shall convert all billing data into a unified currency (e.g., USD) based on daily exchange rates.

4.3 Cost Prediction

FR3: Cost Prediction *(Subsystem: Application Service - Forecast module)*

- **FR3.1:** The system shall apply machine learning time-series models to forecast future cloud costs.
  - **FR3.1.1** The system shall generate cost forecasts for the next 30, 60, and 90 days.
  - **FR3.1.2** The system shall account for historical seasonality (e.g., end-of-month processing spikes or holiday traffic scaling) in its predictions.
- **FR3.2:** The system shall identify and provide forward-looking insights into spending trends.

4.4 Optimization Recommendations

FR4: Optimization Recommendations *(Subsystem: Application Service - Optimization module)*

- **FR4.1:** The system shall detect inefficiencies in infrastructure utilization.
  - **FR4.1.1** The system shall identify ”zombie” or stale assets.
  - **FR4.1.2** The system shall identify overprovisioned compute resources (e.g., instances running consistently below 10% CPU utilization).

4.5 Web-Based Dashboard

FR5: Web-Based Dashboard *(Subsystem: Presentation Service - Dashboard module)*

- **FR5.1:** The system shall provide a unified, interactive web interface to visualize usage metrics.
  - **FR5.1.1** The system shall allow users to filter dashboard views by cloud provider, region, service type, and custom resource tags.
  - **FR5.1.2** The system shall support the creation of custom, user-defined dashboard layouts using drag-and-drop widgets.

4.6 User and Account Management

FR6: User and Account Management *(Subsystem: Application Service - Authentication module)*

- **FR6.1:** The system shall allow users to connect multiple cloud accounts and manage environments.
- **FR6.2:** The system shall control access permissions and manage user identities securely.

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5 | Non-Functional Requirements

5.1 Performance

NFR1.1: The system shall retrieve historical metrics for visualization with a p95 request latency of less than 500 ms to ensure a responsive dashboard.

NFR1.2: After the initial billing forecast, the system shall retrieve subsequent forecasts within a 30-minute period with a p95 request latency of less than 200 ms to ensure responsive billing forecast pages.

NFR1.3: The usage ingestion pipeline shall normalize and persist cloud usage metric records at a throughput of at least 100 records per second under normal operating conditions.

NFR1.4: The billing ingestion pipeline shall normalize and persist cloud billing records at a throughput of at least 100 records per second under normal operating conditions.

5.2 Scalability

NFR2.1: The system should successfully support 100 concurrent users, representing a stress-level load for CloudSherpa, for a period of 1 hour without experiencing more than a 10% degradation in the response times of the below endpoints compared to regular load.

NFR2.1.1: Historical Metric Fetch Endpoints.

NFR2.1.2: Billing Forecasting Endpoints.

NFR2.1.3: Usage Forecasting Endpoint.

5.3 Security

NFR3.1: All sensitive user data and stored cloud provider credentials transmitted across the system's external network boundary shall be protected using TLS 1.3, with TLS terminated at the system ingress.

NFR3.2: NFR3.2: CloudSherpa shall access client AWS, Azure, and GCP billing APIs using least-privelage permissions.

NFR3.3: User metrics must remain separated such that no user gains access to resource details or cost and metric information from accounts not belonging to that user, as this information is potentially highly sensitive to individuals or corporations.

NFR3.4: The system shall have no known high or critical vulnerabilities in third-party dependencies on the main branch.

5.4 Reliability

NFR4.1: In the event of a third-party cloud provider API failure or timeout (e.g., an AWS CloudWatch outage), the ingestion pipeline shall automatically retry the connection and keep track of the failure period, ensuring 0% duplication or metric loss once the provider is back online.

5.5 Maintainability

NFR5.1: The codebase must maintain a minimum of 80% automated test coverage across all core components.

NFR5.2: Static Analysis Should show zero mainatainability issues for CloudSherpa services

5.6 Usability

NFR6.1: A new non-technical user should be able to complete core tasks (e.g., viewing forecasts or checking anomaly alerts) within 5 minutes of first using the system, achieving at least 85% user satisfaction during usability testing.

NFR6.2: The CloudSherpa dashboard shall render correctly on modern web browsers to ensure plain-language summaries remain highly accessible.

5.7 Availability

Scope of Availability

Availability requirements apply from **one week before each COS301 demonstration** until **five days after the demonstration**, due to the limitations of the available free-tier infrastructure.

NFR7.1: The dashboard shall maintain an availability of at least 95% during the defined availability period.

NFR7.2: The application service, responsible for all functionality excluding ingestion and forecasting, shall maintain an availability of at least 95% during the defined availability period.

NFR7.3: The ingestion service, responsible for ingesting cloud usage and billing data, shall maintain an availability of at least 95% during the defined availability period.

NFR7.4: The intelligence service, responsible for forecasting functionality, shall maintain an availability of at least 95% during the defined availability period.

6 Domain Model

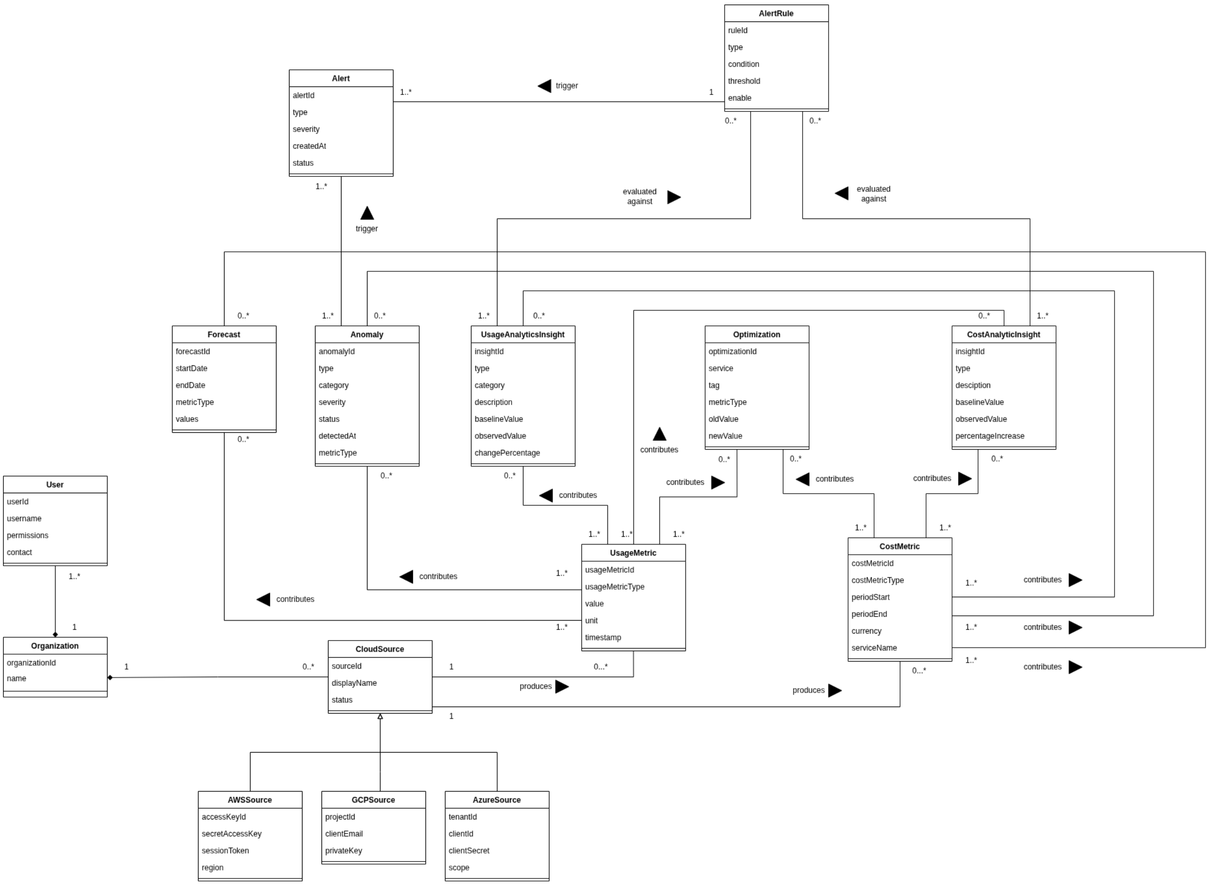

Figure 17: CloudSherpa Domain Model